BX-240 N-Channel Enhancement Mode Power MOSFET, 30 V, 42 A

Features

- V_{DS} = 30 V, I_D = 42 A
- $R_{DS(on)}$ < 13 mΩ @ V_{GS} = 10 V
- Low gate charge (typ. 26 nC)
- 100% avalanche tested
- RoHS compliant, halogen free

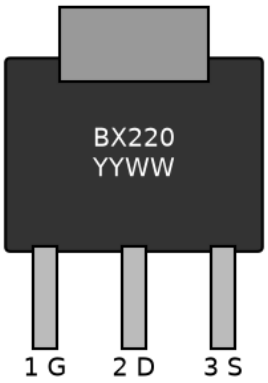
Applications

- DC/DC converters
- Motor drives
- Load switching

Description

The BX-240 uses an advanced trench process to provide excellent $R_{DS(on)}$ with low gate charge. It is suitable for high-frequency switching and synchronous rectification. The device is offered in a TO-252 (DPAK) package and is qualified according to AEC-Q101.

TO-252 (DPAK) - Top View



Tab = Drain | Marking code: B22

Absolute Maximum Ratings (T_A = 25 °C unless otherwise noted)

Parameter	Symbol	Value	Unit
Drain-Source Voltage	VDS	30	V
Gate-Source Voltage	VGS	±20	V
Continuous Drain Current, TC = 25 °C	ID	42	A
Continuous Drain Current, TC = 100 °C	ID	29.6	A
Pulsed Drain Current (1)	IDM	98	A
Single Pulse Avalanche Energy (2)	EAS	39	mJ
Power Dissipation, TC = 25 °C	PD	41	W
Operating Junction and Storage Temperature	TJ, TSTG	-55 to +175	°C

Electrical Characteristics (T_J = 25 °C unless otherwise noted)

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Static Characteristics						
Drain-Source Breakdown Voltage	V(BR)DSS	VGS = 0 V, ID = 250 µA	30	-	-	V
Zero Gate Voltage Drain Current	IDSS	VDS = 30 V, VGS = 0 V	-	-	1	µA
Gate-Body Leakage Current	IGSS	VGS = ±20 V, VDS = 0 V	-	-	±100	nA
Gate Threshold Voltage	VGS(th)	VDS = VGS, ID = 250 µA	1.0	1.6	2.2	V
Static Drain-Source On-Resistance	RDS(on)	VGS = 10 V, ID = 11 A	-	10	13	mOhm
		VGS = 4.5 V, ID = 8 A	-	13	17	mOhm
Forward Transconductance	gFS	VDS = 5 V, ID = 11 A	-	47	-	S
Dynamic Characteristics						
Input Capacitance	Ciss	VDS = 30 V, VGS = 0 V,	-	2140	-	pF
Output Capacitance	Coss	f = 1 MHz	-	278	-	pF
Reverse Transfer Capacitance	Crss		-	79	-	pF
Total Gate Charge	Qg	VDS = 30 V, ID = 11 A,	-	26	39	nC
Gate-Source Charge	Qgs	VGS = 10 V	-	6.7	-	nC
Gate-Drain Charge	Qgd		-	7.9	-	nC
Switching Characteristics						
Turn-On Delay Time	td(on)	VDD = 30 V, ID = 11 A,	-	6	-	ns

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Rise Time	tr	VGS = 10 V, RG = 3.3 Ohm	-	10	-	ns
Turn-Off Delay Time	td(off)		-	35	-	ns
Fall Time	tf		-	5	-	ns
Drain-Source Diode Characteristics						
Diode Forward Voltage	VSD	VGS = 0 V, IS = 11 A	-	0.81	1.1	V
Reverse Recovery Time	trr	IF = 11 A, di/dt = 100 A/µs	-	38	-	ns
Reverse Recovery Charge	Qrr		-	41	-	nC

Notes: (1) Repetitive rating; pulse width limited by maximum junction temperature. (2) Starting TJ = 25 °C, L = 0.5 mH, IAS = 14.7 A.

Typical Characteristics

Fig. 1 Output Characteristics

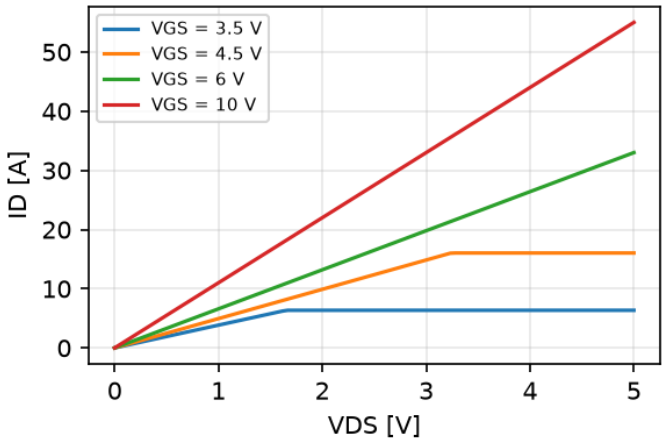
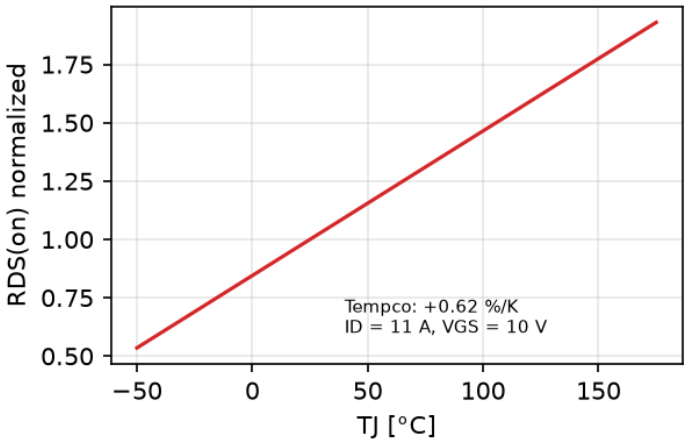


Fig. 2 On-Resistance vs. Temperature



Ordering Information

Part Number	Package	Marking	Packing	MOQ
BX-240-DPK-T	TO-252	B40	Tape & Reel, 2500 pcs	2500
BX-240-DPK-B	TO-252	B40	Tube, 75 pcs	750